

Modbus communication protocol (Single phase)

Note: If adopt Modbus, the controller should be unplug to the PCB.

Frame transfer: RTU mode, 8 data bits, 1 stop bit, no check bit, baud rate 9600

Controller is Master, PCB is Serve. Serve address is 0x01.

Code	Description
03 (03H)	Read holding registers
06 (06H)	Write single registers
16 (10H)	Write multiple registers

Modbus-RTU mode:

Device address	Code	Data layout	CRC check L	CRC check H
8bits	8bits	N*8bits	8bits	8bits

Master read data from Serve:

Serve address	Code	Data address 0x0001	Data qty: 1	CRC data: CAD5
0x01	0x03	00 01 (MSB+LSB)	00 01 (MSB+LSB)	D5 CA

Serve response:

Serve address	Code	Data bit qty	Data: 0x0017	CRC data: 4AF8
0x01	0x03	0x02	00 17 (MSB+LSB)	F8 4A

Master write single register:

Serve address	Code	Data address 0x0001	Data: 0x0001	CRC data: HL
0x01	0x06	00 01 (MSB+LSB)	00 01 (MSB+LSB)	CRCL CRCH

Serve response:

Serve address	Code	Data address 0x0001	Data: 0x0001	CRC data: HL
0x01	0x06	00 01 (MSB+LSB)	00 01 (MSB+LSB)	CRCL CRCH

Master write multiple registers:

Serve address	Code	Data address 0x0001	Data qty: 0x0009	Data bit qty	Data field	CRC data: HL
0x01	0x10	00 01 (MSB+LSB)	00 09 (MSB+LSB)	00 12 (MSB+LSB)		CRCL CRCH

Serve response:

Serve address	Code	Data address 0x0001	Data qty: 0x0009	CRC data: HL
0x01	0x10	00 01 (MSB+LSB)	00 09 (MSB+LSB)	CRCL CRCH

Detailed setting:

Description	Address	Read/ Write	Remark
Situation	30	Read/ Write	0=Turn off 1=Eco heating (Auto mode should refer to 203) 2=Boost heating 4=Cooling
Auto mode	203	Read/ Write	0=No auto mode 1=Auto mode

Modbus communication protocol (Tri-phase)

Communication adopts RS485 bus, asynchronous serial signal has 1 start bit, 8 data bits, 1 end bit, no parity check, baud rate 9600bps.

Comply with the standard MODBUS RTU protocol, 16-bit data structure, 16-bit CRC check low byte is in front of high byte.

The external control signal is Master. The controller is Serve.

Command 03H (query 1 or more registers)

Send command: [device address]+[command number 03H] + [start register address high 8 bits] + [low 8 bits] + [read register number high 8 bits] + [low 8 bits] + [CRC calibration lower 8 bits of check] + [upper 8 bits of CRC check]

Device response: [device address] + [command number 03H] + [number of bytes returned] + [data 1] + [data 2] +...+ [data n] + [low 8 bits of CRC checksum]] + [higher 8 bits of CRC check]

Command 06H (modify a single register)

Send command: [device address] + [command number 06H] + [high 8 bits of register address to be placed] + [low 8 bits] + [high 8 bits of lower data] + [low 8 bits] + [CRC The lower 8 bits of the check] + [the upper 8 bits of the CRC check]

Device response: If the command sent by the computer is successfully returned, otherwise no response

Command 10H (modify multiple registers)

Send command: [device address] + [command number 10H] + [start register address high 8 bits] + [low 8 bits] + [register number high 8 bits] + [low 8 bits] + [register bytes] + [Data 1 upper 8 bits] + [Lower 8 bits] +.... + [Data N upper 8 bits] + [Lower 8 bits] + [Lower 8 bits of CRC check] + [Upper 8 bits of CRC check]

Device response: [device address] + [command number 10H] + [start register address high 8 bits] + [low 8 bits] + [register number high 8 bits] + [low 8 bits] + [CRC check low 8 bits] + [higher 8 bits of CRC check]

Detailed setting:

Description	Address	Read/ Write	Remark
Situation	0x003F	Read/ Write	Bite 0: Turn On/OFF
Mode	0x0040	Read/ Write	Bite 0: 1: Heating 2: Cooling 7: Auto Bite 4: 0: Without Boost mode, 1: With Boost mode